



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to Sri Vinayaga Enterprises

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

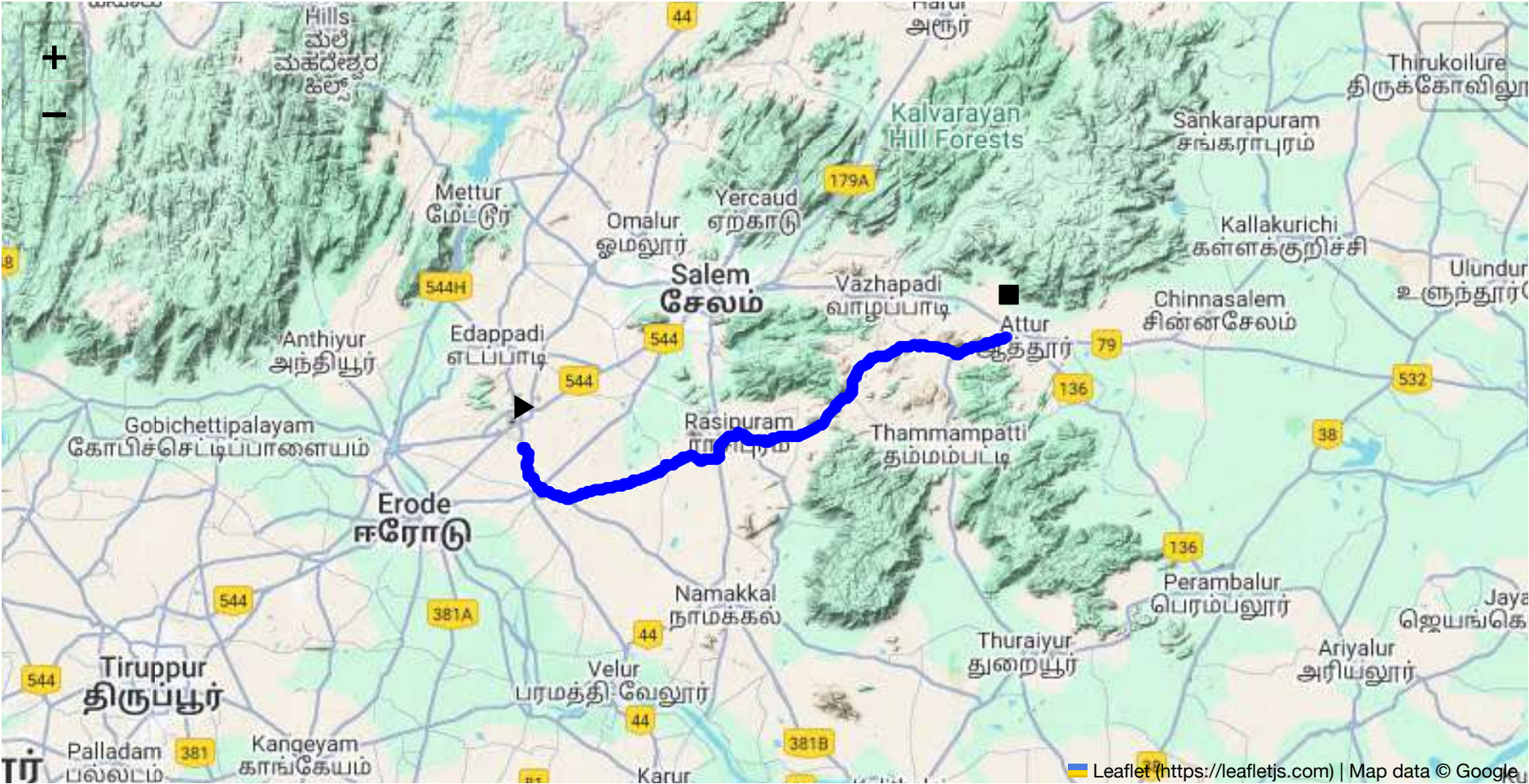
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 97.20 km**  
**Estimated Duration: 2.2 hours**  
**Adjusted Duration (Heavy Vehicle): 2.7 hours**

Start: (11.4381, 77.8734)  
End: (11.5958249, 78.5702078)



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from Sangagiri to Attur via Namakkal, Kavundampalaiyam, Rasipuram, and Koneripatti covers approximately 97.20 kilometers. The primary roads involved are SH 79 and local roads connecting the various towns and villages. The journey takes around 2.15 hours for heavy vehicles carrying hazardous materials, indicating a route that traverses both state highways and local roads.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather with hot summers, moderate monsoons, and mild winters. The climate along this route is typically dry but can become hazardous during the monsoon season (June to September) due to heavy rainfall, leading to slippery roads and potential flooding. Summers (April to June) can cause asphalt roads to become quite hot, posing tire safety issues.

### 3. Analysis of Traffic Patterns

The road from Sangagiri to Namakkal typically sees moderate traffic. Namakkal, being a busy junction, experiences increased congestion, especially during the morning (8:00-10:00 AM) and evening (5:00-7:00 PM) peak hours. Rasipuram and Attur are less congested but can experience local traffic snarls due to markets or local events.

### 4. Assessment of Road Quality and Infrastructure

The SH 79 is generally well-maintained, but some local roads may have potholes or uneven surfaces, particularly near villages and intersections. Periodic maintenance could cause temporary diversions or single-lane traffic.

## 5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or emergencies, alternative routes via the Namakkal-Trichy highway or the Salem-Tiruchirappalli highway can be considered. These routes, although longer, are major connectivity roads and have more reliable infrastructure.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations for the transport of hazardous materials, such as speed limits, restricted hours (usually avoiding night travel in certain areas), and requiring specific permits. Compliance with these rules is crucial to avoid legal and safety issues.

## 7. Overview of Historical Incidents

While specific historical data on incidents involving heavy vehicles or hazardous materials on this route may not be readily available, Tamil Nadu highways occasionally report accidents due to driver negligence or mechanical failure. Regular maintenance of vehicles and adherence to safety norms significantly reduce the risk.

## 8. Environmental Considerations and Sensitive Areas

The route passes through both urban and semi-urban areas, with some roads running close to water bodies and agricultural fields. Care should be taken to prevent hazardous material spills, especially near these sensitive zones, to avoid environmental contamination.

## 9. Analysis of Communication Coverage

The route generally has good mobile network coverage, although certain rural stretches may experience brief dead zones. It is advisable to have alternate communication methods or devices during travel.

## 10. Estimated Emergency Response Times

Emergency response times can vary. In urban and suburban stretches like Namakkal and Rasipuram, response times may be quick (10-20 minutes). However, in more rural areas between these towns, it might take longer (30 minutes to an hour) due to lesser accessibility.

## 11. An Overall Summary of Risk Assessment

Overall, the route poses moderate risks primarily due to traffic congestion, road quality issues, and potential weather impacts during the monsoon season. Adhering to local regulations, routine vehicle checks, and being prepared for communication disruptions will mitigate these risks. Drivers should also prepare for longer travel times during peak traffic hours and monsoon periods.

This risk assessment emphasizes the importance of planning, route familiarity, and preparedness in ensuring a safe journey for heavy vehicles transporting hazardous materials.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates            | Speed Limit |
|----|------------|------------|------------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480     | 15 KM/Hr    |
| 1  | U-Turn     | High       | 11.4535263, 78.1964113 | 10 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345     | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44019, 77.87547     | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.39685, 77.88861     | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.38376, 77.89914     | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.38342, 77.89975     | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.38295, 77.90001     | 30 KM/Hr    |
| 8  | Turn       | High       | 11.38186, 77.90258     | 15 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.37841, 77.90140     | 10 KM/Hr    |
| 10 | Turn       | Medium     | 11.36754, 77.93919     | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.41354, 78.08075     | 30 KM/Hr    |
| 12 | Turn       | High       | 11.41447, 78.08063     | 15 KM/Hr    |
| 13 | Turn       | High       | 11.41475, 78.08087     | 15 KM/Hr    |
| 14 | Turn       | High       | 11.42964, 78.11587     | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.42447, 78.15264     | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.42611, 78.15532     | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.42651, 78.15557     | 30 KM/Hr    |
| 18 | Turn       | High       | 11.44521, 78.15833     | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.45566, 78.17361     | 30 KM/Hr    |
| 20 | Turn       | High       | 11.45587, 78.18057     | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.45760, 78.18140     | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.45990, 78.18413     | 30 KM/Hr    |
| 23 | Blind Spot | Blind Spot | 11.46035, 78.18534     | 10 KM/Hr    |
| 24 | Turn       | Medium     | 11.45929, 78.18544     | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.45895, 78.18572     | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.45856, 78.18775     | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.45824, 78.18811     | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.45764, 78.19208     | 30 KM/Hr    |
| 29 | Blind Spot | Blind Spot | 11.45100, 78.20385     | 10 KM/Hr    |
| 30 | Turn       | High       | 11.45084, 78.20378     | 15 KM/Hr    |
| 31 | Blind Spot | Blind Spot | 11.45354, 78.19642     | 10 KM/Hr    |
| 32 | Turn       | Medium     | 11.45117, 78.20024     | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.44993, 78.22152     | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.44951, 78.22263     | 30 KM/Hr    |
| 35 | Turn       | High       | 11.44797, 78.22314     | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 36 | Turn      | Medium     | 11.44756, 78.22414 | 30 KM/Hr    |
| 37 | Turn      | Medium     | 11.45353, 78.23584 | 30 KM/Hr    |
| 38 | Turn      | Medium     | 11.49617, 78.32335 | 30 KM/Hr    |
| 39 | Turn      | Medium     | 11.49835, 78.32417 | 30 KM/Hr    |
| 40 | Turn      | Medium     | 11.50506, 78.33087 | 30 KM/Hr    |
| 41 | Turn      | Medium     | 11.50865, 78.33966 | 30 KM/Hr    |
| 42 | Turn      | Medium     | 11.51183, 78.34416 | 30 KM/Hr    |
| 43 | Turn      | Medium     | 11.51870, 78.34816 | 30 KM/Hr    |
| 44 | Turn      | Medium     | 11.54842, 78.35627 | 30 KM/Hr    |
| 45 | Turn      | Medium     | 11.56719, 78.37813 | 30 KM/Hr    |
| 46 | Turn      | Medium     | 11.56940, 78.39856 | 30 KM/Hr    |
| 47 | Turn      | Medium     | 11.57829, 78.51245 | 30 KM/Hr    |
| 48 | Turn      | Medium     | 11.58679, 78.54553 | 30 KM/Hr    |
| 49 | Turn      | Medium     | 11.59108, 78.55637 | 30 KM/Hr    |
| 50 | Turn      | Medium     | 11.59253, 78.55778 | 30 KM/Hr    |

### Emergency Locations

|    | type     | name                                      | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital          | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode           | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                            | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 14 | hospital | Sudha Hospital                            | 11.4555773, 78.1760125 | 30 km/h     | Medium     |
| 15 | hospital | Government Hospital Blood Bank, Rasipuram | 11.4612082, 78.1849249 | 30 km/h     | Medium     |
| 16 | hospital | Government Hospital, Rasipuram            | 11.4612987, 78.1851369 | 30 km/h     | Medium     |
| 17 | hospital | M.R. Hospital, Namakkal                   | 11.4613462, 78.1861702 | 30 km/h     | Medium     |
| 18 | hospital | dr.thangavel dch hospital                 | 11.4619955, 78.1849181 | 30 km/h     | Medium     |
| 19 | police   | Police Station Rasipuram                  | 11.4617825, 78.1847411 | 30 km/h     | Medium     |

|    | type     | name                                | coordinates               | speed_limit | risk_level |
|----|----------|-------------------------------------|---------------------------|-------------|------------|
| 20 | hospital | surya hospital                      | 11.4622163,<br>78.1840544 | 30 km/h     | Medium     |
| 21 | hospital | Rasi Hospital                       | 11.4618368,<br>78.1855695 | 30 km/h     | Medium     |
| 22 | hospital | R K Eye Care Center                 | 11.4589079,<br>78.1913596 | 30 km/h     | Medium     |
| 23 | hospital | Manickkam Hospital                  | 11.4568944,<br>78.1926061 | 30 km/h     | Medium     |
| 24 | hospital | Snathi Hospital                     | 11.4564245,<br>78.1932485 | 30 km/h     | Medium     |
| 26 | hospital | Sugam Hospital,Rasipuram            | 11.4553007,<br>78.1936221 | 30 km/h     | Medium     |
| 27 | hospital | Government Hospital, Namagiripettai | 11.4569484,<br>78.265263  | 30 km/h     | Medium     |
| 28 | clinic   | Dr. Saravanakumar Clinic            | 11.5842002,<br>78.4441223 | 30 km/h     | Medium     |
| 29 | hospital | Government Hospital                 | 11.572472, 78.497407      | 30 km/h     | Medium     |
| 31 | hospital | Abi Hospital                        | 11.59319, 78.560129       | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                            | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                    | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                   | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை             | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 6  | marketplace | Farmer Market                                   | 11.4146849,<br>78.0805463 | 30 km/h     | Medium     |
| 7  | school      | Government Higher Secondary School              | 11.4150045,<br>78.0838172 | 30 km/h     | Medium     |
| 8  | school      | Government school                               | 11.4248027,<br>78.1323902 | 30 km/h     | Medium     |
| 9  | college     | Paavai Arts And Science College For Women       | 11.4262443,<br>78.1546987 | 30 km/h     | Medium     |
| 10 | school      | SRV EXCEL Matriculation Higher Secondary School | 11.4299679,<br>78.152616  | 30 km/h     | Medium     |
| 11 | college     | Thiruvalluvar Govt arts college                 | 11.4449352,<br>78.1595562 | 30 km/h     | Medium     |

|    | type   | name                       | coordinates               | speed_limit | risk_level |
|----|--------|----------------------------|---------------------------|-------------|------------|
| 12 | school | Venkateswara School        | 11.4468258,<br>78.1595761 | 30 km/h     | Medium     |
| 13 | school | Venkateswara Middle school | 11.4472063,<br>78.1596191 | 30 km/h     | Medium     |
| 25 | school | Govt. Hr. Sec. School      | 11.4585324,<br>78.1933666 | 30 km/h     | Medium     |
| 30 | school | Rasi Matric HR Sec School  | 11.5760027,<br>78.5062808 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.38395, 77.89480



Risk Type: U-Turn  
Risk Level: High  
Speed Limit: 10 KM/Hr  
Coordinates: 11.4535263, 78.1964113



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44019, 77.87547



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39685, 77.88861



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38376, 77.89914



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38342, 77.89975



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38295, 77.90001



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.38186, 77.90258



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.37841, 77.90140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.36754, 77.93919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.41354, 78.08075



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41447, 78.08063



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41475, 78.08087



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.42964, 78.11587



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42447, 78.15264



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42611, 78.15532



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42651, 78.15557



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44521, 78.15833



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45566, 78.17361



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45587, 78.18057



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45760, 78.18140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45990, 78.18413



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.46035, 78.18534



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45929, 78.18544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45895, 78.18572



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45856, 78.18775



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45824, 78.18811



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45764, 78.19208



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45100, 78.20385



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45084, 78.20378



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45354, 78.19642



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45117, 78.20024



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44993, 78.22152



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44951, 78.22263



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44797, 78.22314



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44756, 78.22414



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45353, 78.23584



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49617, 78.32335



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49835, 78.32417



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50506, 78.33087



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50865, 78.33966



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51183, 78.34416



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51870, 78.34816



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.54842, 78.35627



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56719, 78.37813



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56940, 78.39856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57829, 78.51245



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58679, 78.54553

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